A2.2 Cell structure (HL)

NAME	DATE	TOTAL
		_____ / 30

About 30 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

7 MARKS

1 **STATE** three features of mitochondria that suggest they were once free-living bacteria. [A2.2.12] [3]

2 **NAME** the group of bacteria whose DNA is most similar to chloroplast DNA. [A2.2.12] [1]

3 **DEFINE** proteome. [A2.2.13] [1]

4 **STATE** two advantages of being multicellular. [A2.2.14] [2]

Section B: Understanding

11 MARKS

5 **OUTLINE** how chloroplasts are thought to have arisen by endosymbiosis. [A2.2.12] [3]

6 **EXPLAIN** why mitochondria have two membranes. [A2.2.12] [2]

7 **EXPLAIN** how cells with identical genomes become specialized for different functions. [A2.2.13] [3]

8 EXPLAIN why multicellularity is thought to have evolved many times. [A2.2.14] [2]

9 SUGGEST one disadvantage to a cell of being highly specialized. [A2.2.13] [1]

Section C: Working with evidence

8 MARKS

All data in this section are invented for practice. The patterns are realistic.

10 [A2.2.12]

Liver cells were broken open and separated into two fractions: the cytoplasm (without organelles) and isolated mitochondria. Each fraction was given radioactive amino acids, with or without an antibiotic. Protein synthesis was measured as a percentage of the rate with no antibiotic.

ANTIBIOTIC	CYTOPLASM / %	ISOLATED MITOCHONDRIA / %
CHLORAMPHENICOL	97	8
CYCLOHEXIMIDE	4	95

a IDENTIFY, with a reason, which antibiotic blocks 80S ribosomes. [1]

b COMPARE the effects of the two antibiotics on the two fractions. [2]

c EXPLAIN how these results support the endosymbiotic theory. [2]

d Chloramphenicol is used to treat some bacterial infections. SUGGEST why high doses can harm human tissues that use a lot of energy. [2]

e STATE the purpose of measuring protein synthesis with no antibiotic. [1]

Section D: Longer answer

4 MARKS

11 DISCUSS how strongly the evidence supports an endosymbiotic origin of chloroplasts. [A2.2.12] [4]